

# Experimentation, Causal Inference, Metrics, Modeling, and MLOps (Everything Explained)

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# Agenda

- 1 A/B Testing: Core Concepts
- 2 Hypothesis Testing & Power
- 3 Multiple Testing & Sequential Designs
- 4 Data Quality & Integrity
- 5 Interference & Network Effects
- 6 When A/B Isn't Feasible
- 7 Modeling & Evaluation
- 8 Monitoring & Drift
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- 15 Appendix: Formulas & Checks

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# A/B Testing (Controlled Experiments)

**Definition:** Randomly split units into Control (A) and Treatment (B) and compare outcomes.

**Why:** Randomization balances observed/unobserved factors  $\Rightarrow$  causal attribution.

**Units of Randomization:** user, session, cluster/geo.

**Key rule:** use the smallest unit that avoids *interference*.

- Stable hashing for assignment (e.g., `hash(user_id) mod K`).
- Stratification (blocking) by country/device to reduce variance.
- Exposure integrity: only eligible, actually-exposed users in analysis.

- **Primary metric:** single pre-declared decision metric (e.g., D7 retention).
- **Secondary metrics:** additional success indicators (adoption, time-to-value).
- **Guardrails:** safety metrics that must not degrade (e.g., p95 latency, crash-free sessions).
- **KPI:** Key Performance Indicator; connects work to OKRs (Objectives & Key Results).

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- **Null ( $H_0$ ):** no effect; **Alternative ( $H_1$ ):** effect exists.
- **p-value:** probability of stats as extreme as observed, if  $H_0$  true.
- $\alpha$  (**Type I error**): false positive rate (commonly 0.05).
- $\beta$  (**Type II error**): false negative rate; **Power** =  $1 - \beta$ .
- **Confidence Interval (CI):** Range that would contain the true effect in repeated samples (under the model).

# Sample Size & MDE (Minimum Detectable Effect)

Two-proportion sample size per arm (approx.):

$$n \approx \frac{2 \bar{p}(1 - \bar{p}) (z_{1-\alpha/2} + z_{1-\beta})^2}{\text{MDE}^2}$$

- $\bar{p}$ : baseline rate (e.g., 0.12);  $z_{1-\alpha/2} \approx 1.96$  for  $\alpha = 0.05$ ;  $z_{1-\beta} \approx 0.84$  for 80% power.
- **Duration:** days  $\approx \frac{n}{\text{eligible users/day/arm}}$ , then round to full weeks to cover seasonality.



# Variance Reduction: CUPED

**CUPED** = Controlled Experiments Using Pre-Experiment Data.

- Use pre-period covariate  $X$  correlated with outcome  $Y$ .

**Adjustment:**  $Y^* = Y - \theta (X - \mathbb{E}[X])$ ,  $\theta = \frac{\text{Cov}(Y, X)}{\text{Var}(X)}$

**Variance factor:**  $\text{Var}(Y^*) \approx (1 - R^2) \text{Var}(Y)$  where  $R^2$  comes from regressing  $Y$  on  $X$ .

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# Multiple Testing

- Testing many variants/slices inflates false positives.
- **FWER** (Family-Wise Error Rate):  $\text{Prob}(\geq 1 \text{ false positive})$ .
- **FDR** (False Discovery Rate): Expected fraction of false among declared positives.
- **Controls**: Holm–Bonferroni (FWER, more powerful than Bonferroni); Benjamini–Hochberg (FDR).

# Sequential Testing (Interim Looks)

- **Peeking** inflates Type I error.
- **Fixed-horizon:** Decide once at the end.
- **Alpha-spending:** e.g., O'Brien–Fleming boundaries allocate  $\alpha$  across interim looks with conservative early thresholds.

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## SRM (Sample Ratio Mismatch)

**Definition:** Observed allocation differs from expected (e.g., 50/50 planned, 54/46 observed).

**Why it matters:** Indicates routing/eligibility/bot issues that can bias estimates.

**Detection:**  $\chi^2$  goodness-of-fit on counts; alert if  $p < 0.001$  sustained.

# Exposure Integrity & Instrumentation

- Idempotent events with keys; link exposure  $\rightarrow$  outcome.
- Normalize time zones (store in UTC); handle late events with watermarks.
- Exclude bots/internal traffic; audit coverage and eligibility.

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# When Independence Fails

- **Interference:** One unit's treatment affects another's outcome (social features, shared infra).
- **Mitigations:** cluster randomization (geo/store), switchback experiments, measure spillovers.
- Use cluster-robust standard errors in inference.

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# Difference-in-Differences (DiD)

- Compare before/after changes between treated and control groups.
- **Assumption:** Parallel trends.
- **Good practice:** Event-study plots; cluster-robust SEs; wild bootstrap if few clusters.

- **Synthetic Control:** Weighted donor pool mimics treated pre-period; validate via placebo-in-space/time.
- **RDD:** Treatment at threshold; check manipulation (McCrary), estimate locally with optimal bandwidth.
- **IV:** Instrument  $Z$  affects treatment  $T$  but not outcome  $Y$  directly; requires relevance, exogeneity, exclusion.

## Propensity Scores (PSM/PSW)

- Model  $P(T = 1 \mid X)$  to match/weight units and balance covariates.
- **Diagnostics:** Standardized Mean Difference (SMD)  $< 0.1$ , overlap, no extreme weights.
- Sensitivity: Rosenbaum bounds for unobserved confounding.

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# Logit/Probit (Binary Models)

- **Logit:**  $\text{logit}(p) = \log \frac{p}{1-p} = \beta_0 + \beta^\top x$ ; odds ratios  $e^{\beta_j}$ .
- **Probit:**  $\Phi^{-1}(p) = \beta_0 + \beta^\top x$ ; similar to logit.
- **Regularization:** L1 (sparsity), L2 (stability/multicollinearity).
- **Calibration:** Reliability plots; Platt scaling or isotonic regression.

- **Random Forest:** bagged trees; robust; limited tuning.
- **GBMs (XGBoost/LightGBM):** sequential trees; strong on tabular data.
- **NNs:** FFN (tabular), CNN (images), RNN/LSTM (sequences). Regularize with dropout/weight decay.



- **Imbalance:** Class weights/focal loss; avoid SMOTE in time series.
- **Metrics:** ROC-AUC, PR-AUC, Precision, Recall, F1, Brier score (probability calibration).
- **Explainability:** Feature importance, PDP/ICE, SHAP (global & local).

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- **Data drift**: feature distribution shifts. **Concept drift**: relationship changes.
- **PSI (Population Stability Index)**: binned divergence; org thresholds (e.g., 0.1, 0.25).
- **KS test**: max CDF distance; sensitive on large  $n$ .

- **Triggers:** retrain on drift thresholds or performance decay.
- **Cadence:** scheduled retrains with backtesting before promotion.
- Track schema checks, latency, decision logs.

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- **CI/CD:** Continuous Integration/Delivery—tests, builds, deploys.
- **Canary:** small % rollout; measure before expand.
- **Shadow:** parallel predictions; no user impact.
- **Feature store:** consistent batch/online features.
- **Model registry:** versions, lineage, approvals.
- **Kill switch:** instant rollback.

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- Start from the decision/question; one message per slide.
- Maximize data-ink ratio (Tufte); remove chart junk.
- Honest axes; colorblind-safe palettes; add context lines/targets.



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# Cohort Retention (SQL Skeleton)

```
WITH installs AS (  
    SELECT user_id, MIN(event_ts) AS install_ts  
    FROM events  
    WHERE event_name = 'install'  
    GROUP BY 1  
)  
,  
activity AS (  
    SELECT e.user_id,  
        DATE_DIFF('day', i.install_ts, e.event_ts) AS dfi  
    FROM events e  
    JOIN installs i USING (user_id)  
    WHERE e.event_name = 'app_open'  
        AND e.event_ts BETWEEN i.install_ts  
                                AND i.install_ts + INTERVAL '28 day'  
)  
,  
dedup AS (  
    SELECT user_id, dfi,  
        ROW_NUMBER() OVER (  
            PARTITION BY user_id, dfi ORDER BY updated_at DESC  
        ) AS rn  
    FROM activity  
)  
SELECT dfi,  
    COUNT(DISTINCT CASE WHEN dfi = 0 THEN user_id END) AS n0,  
    COUNT(DISTINCT CASE WHEN rn = 1 THEN user_id END) AS active,  
    COUNT(DISTINCT CASE WHEN rn = 1 THEN user_id END)::float  
    / NULLIF(COUNT(DISTINCT CASE WHEN dfi = 0 THEN user_id END), 0)  
    AS retention  
FROM dedup  
GROUP BY 1  
ORDER BY 1;
```

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- **KM:** Nonparametric survival  $S(t)$  with censoring.
- **Cox PH:** Hazard model with multiplicative covariate effects; test proportional hazards via Schoenfeld residuals.
- Compare cohorts with log-rank test.

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# Common Pitfalls

- Peeking without correction  $\Rightarrow$  inflated Type I error.
- Ignoring SRM  $\Rightarrow$  biased estimates.
- Leakage from future/post-treatment variables.
- Metric misalignment (optimize clicks vs. retention).
- Multiple testing without correction; Simpson's paradox.
- No post-ship monitoring; regression to the mean.

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# Quick Reference

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|                |                                                         |
|----------------|---------------------------------------------------------|
| A/B            | Control vs. treatment experiment                        |
| KPI            | Key Performance Indicator                               |
| SRM            | Sample Ratio Mismatch                                   |
| MDE            | Minimum Detectable Effect                               |
| CUPED          | Variance reduction using pre-period covariates          |
| FWER           | Family-Wise Error Rate                                  |
| FDR            | False Discovery Rate                                    |
| RCT            | Randomized Controlled Trial                             |
| DiD            | Difference-in-Differences                               |
| RDD            | Regression Discontinuity Design                         |
| IV             | Instrumental Variables                                  |
| PSM/PSW        | Propensity Score Matching/Weighting                     |
| ROC-AUC/PR-AUC | Discrimination summaries under class imbalance          |
| Brier          | Probability calibration error (MSE of probs)            |
| PSI            | Population Stability Index                              |
| KS             | Kolmogorov–Smirnov test                                 |
| p95            | 95th percentile (latency tail)                          |
| PDP/ICE        | Partial Dependence / Individual Conditional Expectation |
| SHAP           | Shapley-based local/global explanations                 |



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- **Two-proportion MDE (given  $n$ ):**  $\text{MDE} \approx \sqrt{\frac{2 \bar{p}(1 - \bar{p})(z_{1-\alpha/2} + z_{1-\beta})^2}{n}}$
- **CUPED:**  $Y^* = Y - \theta(X - \mathbb{E}[X])$ ,  $\theta = \frac{\text{Cov}(Y, X)}{\text{Var}(X)}$
- **BH-FDR:** sort p-values  $p_{(i)}$ , find largest  $k$  with  $p_{(k)} \leq \frac{k}{m}q$ , declare  $1..k$ .
- **Holm:** order p-values; compare  $p_{(i)} \leq \frac{\alpha}{m-i+1}$  sequentially.
- **O'Brien–Fleming (alpha-spending):** conservative early, liberal late boundaries.

- Pre-register primary metric, guardrails, decision rule.
- Powering with realistic MDE; round duration to full cycles.
- Instrumentation dry run; SRM alarms; exposure audits.
- Sensitivity: heterogeneity, alternative tests (Welch/MWU), outliers.
- Post-ship: DiD vs. non-adopters; kill switch; rollback plan.

Questions & Discussion